

Technology Stack

Date	04 July 2026
Team ID	SWTID-2026-2950
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	React.js + Vite, HTML5, CSS3, JavaScript	Provides a fast, responsive, and modern user interface. React enables reusable components and Vite improves development speed and performance.

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	Python 3.11 + FastAPI	FastAPI offers high performance, automatic API documentation, secure REST APIs, and easy integration with AI services.
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	SQLite + SQLAlchemy ORM	SQLite is lightweight and ideal for internship projects, while SQLAlchemy simplifies database operations using ORM.
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	Docker, Google Cloud Platform (GCP)	Docker ensures consistent deployment, and Google Cloud provides scalable and reliable cloud hosting for AI-enabled applications.
5	Version Control & CI/CD	Git, GitHub	Git manages source code versions efficiently, while GitHub enables collaboration, repository management, and continuous integration.

	<i>(e.g., GitHub, GitLab, Jenkins)</i>		
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	Google Gemini API, VS Code, Postman	Google Gemini API generates AI-powered settlement recommendations and negotiation letters. VS Code is used for development and Postman for API testing.